

Solutions: Probability Theory II

1. Consider the p.d.f. $f(x) = 2x$ for $0 \leq x \leq 1$.

(a) Calculate the c.d.f. of $f(x)$.

$$F(x) = x^2$$

(b) Is $f(x)$ a proper p.d.f.?

$$\begin{aligned}\int_0^1 F(x) &= 1 \\ 1^2 - 0^2 &= 1\end{aligned}$$

$f(x)$ is a proper p.d.f.

2. Consider the c.d.f. $G(x) = \frac{1}{9}x^2$ for $0 \leq x \leq 3$.

(a) Calculate the p.d.f. of $G(x)$, $g(x)$.

$$g(x) = \frac{2}{9}x$$

(b) Is $g(x)$ a proper p.d.f.?

$$\begin{aligned}\int_0^3 G(x) &= 1 \\ \frac{1}{9} \times 3^2 - \frac{1}{9} \times 0^2 &= 1\end{aligned}$$

$g(x)$ is a proper p.d.f.

3. Consider the p.d.f. $h(x) = \frac{4}{3}(1 - x^3)$ for $0 < x < 1$. Determine

(a) $\Pr(X < \frac{1}{2})$.

$$\begin{aligned}H(x) &= -\frac{1}{3}x(x^3 - 4) \\ H\left(\frac{1}{2}\right) &\approx 0.65\end{aligned}$$

(b) $\Pr(X > \frac{1}{3})$.

$$1 - H\left(\frac{1}{3}\right) \approx 0.56$$

(c) $\Pr(\frac{1}{4} < X < \frac{3}{4})$.

$$H\left(\frac{3}{4}\right) - H\left(\frac{1}{4}\right) \approx 0.56$$

4. Consider the p.d.f. $k(x) = cx^2$ for $1 \leq x \leq 2$. Determine

(a) Find the value of the constant c .

$$\begin{aligned} \int_1^2 k(x) dx &= 1 \\ \frac{1}{3}cx^3 \Big|_1^2 &= 1 \\ c &= \frac{3}{7} \end{aligned}$$

(b) Find $\Pr(X > \frac{3}{2})$.

$$\begin{aligned} K(x) &= \frac{1}{7}x^3 \\ 1 - K\left(\frac{3}{2}\right) &\approx 0.52 \end{aligned}$$